McCabe-Thiele METHOD

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INTRODUCTION

WHAT IS?

The McCabe-Thiele method is a graphical technique used in distillation column design to determine the **number of theoretical trays** required for separating a binary mixture. It helps in visualizing the equilibrium stages in a distillation column.

WHY WE NEED?

- Provides a stepwise graphical approach to analyze distillation.
- Helps determine the optimal number of trays for a given separation.
- Evaluates the impact of reflux ratio, feed quality, and operating lines on separation efficiency
- We use MATLAB to automate the McCabe-Thiele construction, analyze separation efficiency, and optimize energy consumption in distillation columns.

PRINCIPLES & ASSUMPTIONS

- **Vapor-Liquid Equilibrium**: Each tray reaches equilibrium between vapor and liquid phases.
- **Constant Molar Overflow**: Vapor and liquid flow rates remain constant within each column section.
- **Binary Mixture**: Only two components are considered.
- Graphical Approach: The number of trays is determined by stepping off stages.

APPLICATIONS & IMPORTANCE

INDUSTRIAL

- *Petroleum Industry*: Used in refining crude oil into gasoline, diesel, and other fractions.
- *Pharmaceuticals*: Purification of solvents and active compounds.
- *Alcohol Production*: Used for ethanol-water separation in beverage and biofuel industries.
- *Chemical Processing*: Key in large-scale separation processes for various chemicals.

DISTILLATION COLUMN DESIGN

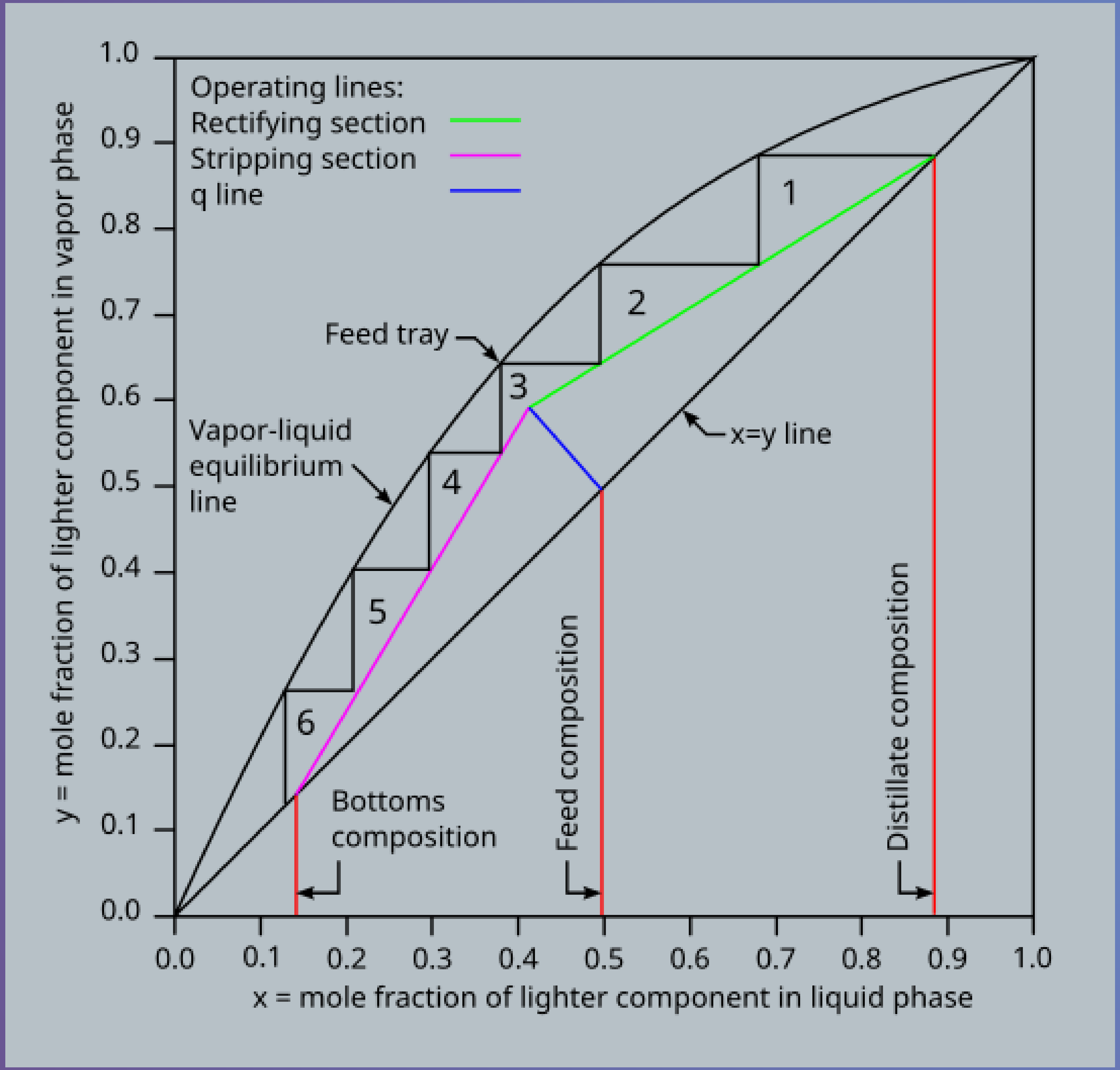
- Determines the number of trays required for effective separation.
- Helps in optimizing the reflux ratio for energy efficiency.
- Provides a visual representation of the distillation process.

MATLAB IMPLIMENTATION

- Automates tray calculations based on equilibrium and mass balance equations.
- Allows dynamic simulations to study how changes in feed composition, reflux ratio, and operating conditions affect separation.

McCabe Diagram

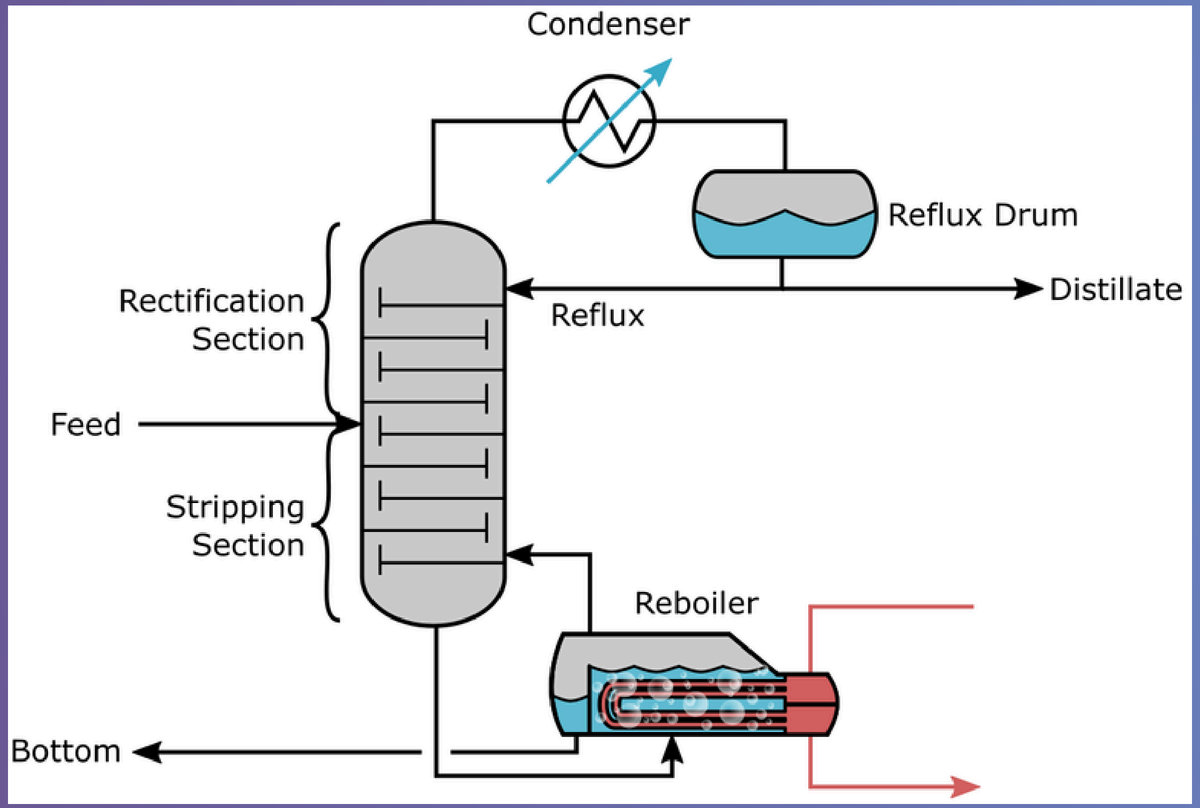
WEEK -1



Key Components of the Diagram

1. Equilibrium Curve – Represents the equilibrium relationship between the vapor and liquid phases, derived from Vapor-Liquid Equilibrium (VLE) data.
2. Operating Lines – Represent mass balances within the distillation column:
   - Rectifying Line: Describes the behavior of the vapor and liquid above the feed tray.
   - Stripping Line: Describes the behavior below the feed tray.
3. 45-Degree Reference Line – A diagonal line (y = x) used as a reference for stepping off theoretical stages.
4. Feed Line (q-Line) – Determines how the feed enters the column based on its phase condition (liquid, vapor, or a mix of both).

TERMILOGY



STRIPPING SECTION

The stripping section is the lower part of the distillation column, below the feed tray. Here, the less volatile component is concentrated as the mixture moves downward, and the vapor phase loses the more volatile component.

$$y_m = \frac{L_s}{V_s} x_m - \frac{x_B}{V_s}(L_s - V_s)$$

Where:

- ym = Vapor composition at stage m
- xm = Liquid composition at stage m
- Ls = Liquid flow rate in the stripping section
- Vs = Vapor flow rate in the stripping section
- xB = Bottoms composition

RECTIFYING SECTION

The rectifying section is the upper part of the distillation column, above the feed tray. In this section, the vapor rising from the column is enriched in the more volatile component due to repeated vapor-liquid equilibrium stages.

$$y_n = \frac{R}{R+1} x_n + \frac{x_D}{R+1}$$

Where:

- yn = Vapor composition at stage n
- xn = Liquid composition at stage n
- R = Reflux ratio
- xD = Distillate composition

REFLUX RATIO

The reflux ratio is the ratio of the liquid returned to the column as reflux to the liquid collected as distillate. It plays a key role in determining the separation efficiency.

$$R = \frac{L}{D}$$

Where:

- R = Reflux ratio
- L = Liquid reflux returned to the column
- D= Distillate collected

STEP BY STEP CONSTRUCTION

WEEK -2

STEP - 1

PLOT THE EQUILIBRIUM CURVE

- THE EQUILIBRIUM CURVE IS OBTAINED USING VLE DATA (EXPERIMENTAL OR ANTOINE'S EQUATION).
- THIS CURVE SHOWS HOW THE LIQUID AND VAPOR COMPOSITIONS RELATE AT EQUILIBRIUM.

STEP - 2

DRAW THE 45-DEGREE REFERENCE LINE

A STRAIGHT $Y = X$ LINE HELPS IN GRAPHICAL STEPPING.

STEP - 3

IDENTIFY THE FEED CONDITION AND DRAW THE Q-LINE

- $Q = 1$ → SATURATED LIQUID (VERTICAL Q-LINE)
- $Q = 0$ → SATURATED VAPOR (HORIZONTAL Q-LINE)
- $0 < Q < 1$ → PARTIALLY VAPORIZED FEED.

STEP - 4

CONSTRUCT THE OPERATING LINES

- RECTIFYING LINE: CALCULATED USING MASS BALANCE EQUATIONS AND THE REFLUX RATIO (R)
- STRIPPING LINE: DETERMINED USING THE INTERSECTION OF THE FEED COMPOSITION AND THE BOTTOM PRODUCT.

STEP - 5

STEPPING OFF STAGES TO DETERMINE TRAY COUNT

- STARTING FROM THE DISTILLATE COMPOSITION, WE STEP BETWEEN THE OPERATING LINES AND EQUILIBRIUM CURVE
- THE NUMBER OF STEPS REPRESENTS THE THEORETICAL NUMBER OF TRAYS REQUIRED

EFFECT OF REFLUX RATIO

WHAT IS REFLUX RATIO ?

- The reflux ratio (R) is the ratio of liquid returned to the column compared to the liquid taken as distillate.
- A higher reflux ratio improves separation but increases energy consumption

Effect of Different Reflux Ratios on the Diagram

1. Low Reflux Ratio (Near Minimum Reflux)
   - Fewer equilibrium stages required.
   - More trays needed → Larger column size.
   - Less energy consumption but poor separation.
2. High Reflux Ratio (Near Total Reflux)
   - More equilibrium stages in the diagram.
   - Fewer trays required → Smaller column.
   - High energy consumption.
3. Optimal Reflux Ratio
   - Balances energy efficiency and separation.
   - MATLAB simulations help find the best R for minimum energy usage.

- MATLAB is used to analyze different reflux ratios by adjusting R in the operating line equations.
- It generates multiple McCabe-Thiele diagrams to compare separation efficiency.
- Dynamic simulations allow optimization of energy consumption.

Advanced McCabe Concepts

Feed Location Optimization

- The position of the feed affects efficiency.
- Proper feed tray selection reduces unnecessary trays and minimizes energy use.
- MATLAB helps visualize feed stage placement by altering q-line placement dynamically.

Non-Ideal Cases & Limitations

- McCabe-Thiele assumes constant molar overflow, which is not valid for all systems.
- For more accurate analysis, rigorous MATLAB simulations (e.g., rate-based models) can be used.

Assignment - 1

T-X-Y & P-X-Y DIAGRAM CONSTRUCTION

OBJECTIVE

- CONSTRUCT A TEMPERATURE-COMPOSITION (T-X-Y) DIAGRAM FOR AN ETHANOL-WATER MIXTURE USING THE ANTOINE EQUATION.
- CONSTRUCT A PRESSURE-COMPOSITION (P-X-Y) DIAGRAM USING RAOULT'S LAW AND EQUILIBRIUM DATA.

STEPS IN MATLAB

1. USE THE ANTOINE EQUATION TO CALCULATE SATURATION PRESSURES AT DIFFERENT TEMPERATURES.

2. APPLY RAOULT'S LAW TO OBTAIN VAPOR-LIQUID EQUILIBRIUM (VLE) DATA.

3. PLOT T-X-Y AND P-X-Y DIAGRAMS IN MATLAB.

KEY TAKEAWAYS

- **UNDERSTANDING THE PHASE BEHAVIOR OF ETHANOL-WATER MIXTURES.**
- **USING MATLAB FOR VLE DATA GENERATION AND VISUALIZATION.**

Assignment - 2

MCCABE FOR BATCH DISTILLATION

OBJECTIVE

- PERFORM MCCABE-THIELE ANALYSIS FOR AN IDEAL BATCH DISTILLATION PROCESS.
- GIVEN AN ETHANOL-WATER MIXTURE, DETERMINE THE NUMBER OF TRAYS REQUIRED.

STEPS IN MATLAB

1. FIT THE GIVEN EXPERIMENTAL DATA TO A FUNCTION USING MATLAB.
2. GENERATE THE EQUILIBRIUM CURVE FROM THE FUNCTION.
3. PLOT OPERATING LINES FOR BATCH DISTILLATION WITHOUT RECTIFYING/STRIPPING SECTIONS.
4. STEP OFF THEORETICAL TRAYS USING MATLAB.

KEY TAKEAWAYS

- UNDERSTANDING MCCABE-THIELE GRAPHICAL STEPPING FOR BATCH DISTILLATION.
- APPLYING MATLAB FOR AUTOMATED TRAY CALCULATION.

Assignment - 3

MCCABE CONSTRUCTION

OBJECTIVE

- CONSTRUCTING THE VAPOR-LIQUID EQUILIBRIUM (VLE) CURVE USING EXPERIMENTAL X,Y DATA.
- FITTING THE EQUILIBRIUM CURVE USING NONLINEAR REGRESSION.
- DEFINING AND PLOTTING OPERATING LINES (RECTIFYING AND STRIPPING SECTIONS).
- PERFORMING STAGE STEPPING CALCULATIONS TO DETERMINE THE NUMBER OF THEORETICAL TRAYS.
- SIMULATING THE FEED INTRODUCTION AND SIDESTREAM REMOVAL, AND ANALYZING THEIR EFFECT ON SEPARATION.

STEPS IN MATLAB

1. DEFINE GIVEN DATA & FIT VLE CURVE
2. MATERIAL BALANCE CALCULATIONS
3. PLOT MCCABE-THIELE DIAGRAM
4. COMPUTE MINIMUM REFLUX RATIO & ACTUAL REFLUX
5. DETERMINE STEPPING-OFF STAGES FOR TRAYS CALCULATION
6. CALCULATE THE NUMBER OF THEORETICAL STAGES

KEY TAKEAWAYS

- MATLAB EFFECTIVELY MODELS BINARY DISTILLATION USING THE MCCABE-THIELE METHOD.
- THE VLE CURVE FITTING HELPS IN ACCURATE GRAPHICAL CONSTRUCTION.
- OPERATING LINES AND REFLUX RATIO PLAY A CRUCIAL ROLE IN DETERMINING SEPARATION EFFICIENCY.
- GRAPHICAL STAGE STEPPING PROVIDES AN ESTIMATE OF THEORETICAL TRAYS REQUIRED.
- THE FEED TRAY POSITION AND SIDESTREAM REMOVAL POINT IMPACT SEPARATION PERFORMANCE.

Assignment - 4

DYNAMIC SIMULATION

OBJECTIVE

- SIMULATE DYNAMIC ETHANOL-WATER DISTILLATION WITH VARYING FEED COMPOSITION AND REFLUX RATIO.
- ENSURE HIGH-PURITY DISTILLATE (≥95%) WHILE MINIMIZING ENERGY CONSUMPTION.
- IMPLEMENT MCCABE-THIELE METHOD AND EXPORT RESULTS TO EXCEL.

STEPS IN MATLAB

- INITIALIZE INPUTS: FEED VARIATION XF(T)X_F(T)XF(T), REFLUX RATIO, TIME STEPS (10 HR, 30 MIN INTERVALS).
- EQUILIBRIUM DATA: USE POLYNOMIAL FITTING FOR VAPOR-LIQUID EQUILIBRIUM.
- SIMULATION LOOP:
- ADJUST REFLUX RATIO DYNAMICALLY.
- COMPUTE DISTILLATE, BOTTOMS COMPOSITIONS, AND ENERGY DUTIES.
- PLOT MCCABE-THIELE, COMPOSITION TRENDS, AND ENERGY CONSUMPTION DYNAMICALLY.
- EXPORT DATA: WRITE RESULTS (COMPOSITIONS, ENERGY DUTIES, STAGES) TO EXCEL.

KEY TAKEAWAYS

- TRADE-OFF: HIGHER PURITY INCREASES ENERGY CONSUMPTION.
- DYNAMIC REFLUX ADJUSTMENT STABILIZES COMPOSITION OVER TIME.
- MCCABE-THIELE METHOD VISUALIZES SEPARATION EFFICIENCY.
- RESULTS AID OPTIMIZATION OF ENERGY-EFFICIENT DISTILLATION.

REFERENCES

1) INTRO:

2)VLE CURVE

3) NPTEL PLAYLIST

4) SUMMARY

5) IN-DEPTH PDF

PROVIDED IN PROJECT

THANK YOU